$\sum a_n, \sum b_n, \frac{a_n}{b_n} \to L \Rightarrow$: si $L > 0$, $\sum a_n, \sum b_n$ de mismo tipo

$L = 0$, $\sum b_n$ cv $\Rightarrow \sum a_n$ cv

a- $\sum_{n\geq 0} \frac{1}{n^2+1}$ b- $\sum_{n\geq 1} \frac{n^2+1}{n^3}$ c- $\sum_{n\geq 1} \frac{\log(n+1)-\log(n)}{10n+1}$

a- $\left(\frac{1}{n^2}\right) = (b_n)$, $(a_n) = \left(\frac{1}{n^2+1}\right)$: $\frac{1/n^2+1}{1/n^2} = \frac{1}{1+1/n^2} \xrightarrow[n\to+\infty]{} 1$

$\sum 1/n^2$ cv. $\sum_{n\geq 1} 1/n$ DV



$x \to 1/x^2$ $2k \to 1/x$

$\int_1^N 1/x^2 = [-1/x]_1^{+\infty} = 0 - -1 = -1$

$\sum \frac{1}{n} \sim \int_1^N 1/x = \ln(N) - \underbrace{\ln(1)}_{0} \xrightarrow[N\to+\infty]{} +\infty$ (Pero $1/n \to 0$)

$\sum_{n\geq 1} \frac{n^2+1}{n^3}$, $\frac{(n^2+1)/n^3}{1/n} = \frac{n(n^2+1)}{n^3} = 1 + 1/n^3 \xrightarrow[n\to+\infty]{} 1 > 0$

$\sum_{n\geq 1} 1/n \geq \int_1^{\infty} 1/x\,dx = \lim_{N\to+\infty} \int_1^N (1/x)\,dx = \lim_{N} \ln(N) - 0$

$\sum \frac{\log(n+1)-\log(n)}{10n+1}$, $\frac{\log\left(\frac{n+1}{n}\right)}{10n+1} = \frac{\log(1+1/n)}{10n+1}$, $\log(1+1/n) = \frac{1}{n} - \frac{1}{2n^2} + o(1/n^2)$

$\log(1+x) = x - x^2/2 + \cdots + (-1)^{n+1}\frac{x^n}{n} + o(x^n)$

$(a_n \sim b_n := \frac{a_n}{b_n} \xrightarrow[n\to+\infty]{} 1)$

$\frac{\log(1+1/n)}{10n+1} = \frac{1/n + o(1/n)}{10n+1} = \frac{1}{10}\cdot\frac{1}{n^2} + o(1/n^2)$

a_n ; $\sim c \cdot 1/n^2$; $\leq 1/n^2$; $= b_n$

$\frac{1}{10}\sum 1/n^2$ cv. cd: $\sum \frac{\log(n+1)-\log(n)}{10n+1}$, $\sum \frac{\log(n)}{n} > \sum \frac{1}{n}$

$a_n = o(n)$ quiere dch

$\frac{a_n}{n} \to 0$,

$a_n = o(b_n)_{n\to+\infty} := \frac{a_n}{b_n} \to 0$

$f(x) = o(x^2)_{x\to 0} := \frac{f(x)}{x^2} \xrightarrow[x\to 0]{} 0$

$f(x) = o(g(x)) := \frac{f(x)}{g(x)} \to 0$

$a_n = O(b_n) := \exists C > 0$ tq

$a_n \leq C \cdot b_n$

$a_n = o(b_n) \Rightarrow a_n = O(b_n)$ $\not\Leftarrow$

$o(a_n) + o(a_n) = o(a_n)$

$\lambda a_n + \mu b_n = o(a_n)$